FML 2026 Challenge
Analysis and Prediction over Bike Sharing Data

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1 Goals

The goal of the challenge is to apply supervised and unsupervised principles and techniques (data analysis, preparation, feature selection, prediction, etc.), seen during the FML course, to a large, real-world, non-trivial dataset. Students work in teams of 5–6, with freedom to organise activities inside each team. There is no single correct solution to the challenge: you are free to explore the data from different angles, free to define the features of your interest, and free to incorporate third-party data into your processing. However, you need to solve the three tasks defined below.

2 Input Dataset

This year, the dataset contains daily usage data of bike sharing systems throughout Europe. The dataset consists of the following tables:

cities.csv: cities with their geographic positions and countries with bike-sharing systems;

bikes.csv: instances of bikes, linked to their types;

bike_types.csv: characteristics of bikes, e.g. human or electric propulsion;

stations.csv: bike sharing stations, their locations and capacities;

station_status.csv: availability of bikes at a given station at a given point in time;

trips.csv: starting and ending times and geographical coordinates of each bike trip.

Your objective is not simply to obtain accurate machine learning models, but to discover meaningful patterns of urban mobility and demonstrate how these patterns can help improve the understanding and operation of European bike-sharing systems.

3 Tasks

You need to provide a solution to all of the tasks below.

3.1 Data Exploration and Preprocessing Task

This task involves data understanding, descriptive statistics, handling missing data in an informed manner, cleaning, transformation, fusion, exploration of the correlation of variables, visualisation of statistics, etc. We will appreciate non-trivial insights gained in this step (e.g. not simply obtainable via the `describe()` method from `pandas`).

3.2 Unsupervised Tasks: Bike Trip and Stations Profiles

The objective of this task is to discover and interpret characteristic patterns in the usage of European bike-sharing systems through unsupervised learning.

You are required to conduct two complementary clustering analyses, focusing on different levels of observation: individual bike trips and bike-sharing stations.

Part A — Clustering Bike Trips

Identify characteristic profiles of bicycle journeys based on their properties and usage patterns.

You are free to define the features and representations you consider relevant, using the provided data and potentially additional external sources.

Your objective is to discover, characterize, and interpret meaningful groups of trips, providing insights into the different ways bike-sharing systems are used.

Part B — Clustering Bike-Sharing Stations

Identify characteristic profiles of bike-sharing stations based on their activity and usage patterns.

You should construct relevant station-level features from the available data and investigate whether different types of stations can be distinguished.

Your objective is to characterize and interpret these profiles and discuss what they reveal about the organization and usage of bike-sharing systems.

Methodological Expectations

For both tasks, you are expected to apply appropriate factorial analysis, clustering, and evaluation techniques.

You are free to select your features, algorithms, and experimental approaches. However, your methodological choices must be justified, and your results must be critically evaluated and interpreted.

Beyond identifying clusters, we expect meaningful insights supported by quantitative evidence, together with a discussion of the limitations of your findings.

You are encouraged to investigate potential relationships between the trip profiles and station profiles discovered in the two analyses.

3.3 Supervised Task: Predict Station Availability

In order to maintain the bike sharing system, bikes regularly need to be moved from overcrowded stations to empty ones. The file `station_status.csv` provides the number of bikes available and the number of free docks at a station at any given time. Your goal is to implement a machine-learning-based solution that helps maintainers predict station usage, so that they can plan the transfer of bikes in advance. Your objective is to develop a supervised regression approach to predict the number of bicycles available for rental at a given station one hour into the future.

Using the provided dataset, and possibly additional external sources, you are expected to construct a suitable learning problem and develop an effective predictive solution.

You are free to select the data, features, preprocessing techniques, algorithms, and evaluation methods you consider appropriate. However, your choices must be scientifically justified.

Before the end of the challenge, we will release an unseen evaluation set, over which you must execute a final evaluation. You are obviously not allowed to retrain your model at this point. In your final submission, you should include the results of your own preliminary evaluations as well as the final one.

4 Deliverables

The submission deadline is: 10 November 2026.

Your submission must include:

- the **annotated, human-readable, professor-friendly, instantly executable** notebook that contains source code, visualisations, as well as clear explanations of your approach;
- a short, synthetic PDF report (3-6 pages) that sums up your approach. It should not include excessive detail nor generic introductory filler text, but it should include motivations of your solution, results, and insights gained.

AI use warning:

- your report, as well as your notebook annotations, must be entirely human-written, AI-generated text is strictly forbidden;

- you are supposed to be able to explain and implement all the source code you are providing, in case of doubt, the professors reserve the right to ask you for specific details;
- in case of a supposed non-compliance with these requirements, the professors reserve the right to attribute a zero score to the entire team.

The best teams will present their work to their peers in class, on 24 November.

5 Evaluation Criteria

Your score for the submission, out of 10, is composed as follows:

- exploration and preprocessing task: 1.5 points;
- unsupervised learning task: 2 points;
- supervised learning task: 2 points;
- final model performance: 0.5 points;
- interpretations and insights: 2 points;
- notebook and report quality: 2 points.

6 Dataset Description

6.1 trips.csv

variable	unit	explanation
bike_id	–	Bike identifier; bikes.id
city_id	–	Rental city; cities.id
time_start	s	Rental start (UTC)
lon_start, lat_start, lon_end, lat_end	°	Start and end coordinates WGS-84
station_id_start, station_id_end	–	Station IDs, NULL if free-floating
battery_start, battery_end	%	State of charge (e-bikes only)
duration	s	Trip duration
distance	m	Great-circle distance (PostGIS ST_Distance)

6.2 cities.csv

variable	unit	explanation
id	–	Primary key
name	–	Name of the bike-sharing scheme
lat, lon	°	Approximate city center
timezone	–	IANA timezone string (e.g. Europe/Paris)
country	–	ISO-3166 alpha-2 country code
return_to_official_only	bool	true if bikes must be left at a station

6.3 bike_types.csv

variable	unit	explanation
id	–	Technical bike type
vehicle_image	–	URL of representative image
name	–	Commercial name
description	–	Free-text description
form_factor	–	regular / cargo
rider_capacity	–	Typical seats (1, 2, . . .)
propulsion_type	–	human / electric
max_range	m	Nominal electric range
battery_capacity	Wh	Battery energy

6.4 bikes.csv

variable	unit	explanation
id	–	Bike identifier
bike_type_id	–	Technical type; bike_types.id
computer_id	–	On-board computer identifier

6.5 stations.csv

variable	unit	explanation
id	–	Station identifier
city_id	–	cities.id
name	–	Human-readable label
app_number	–	Number shown to users
terminal_type	–	Hardware generation (12 values)
place_type	–	Unknown (23 observed values)
bike_racks	–	Regular parking positions
special_racks	–	Charging racks
lon, lat	°	Location

6.6 station_status.csv

variable	unit	explanation
station_id	–	stations.id
time	s	Snapshot timestamp (UTC)
bikes	–	Total bikes currently docked
booked_bikes	–	Bikes reserved by users
bikes_available_to_rent	–	Bikes ready for rental
free_racks	–	Empty regular docks
free_special_racks	–	Empty charging docks
maintenance	bool	true = offline